Use the dataset: https://www.kaggle.com/datasets/kritanjalijain/amazon-reviews/data
Use the Google Word2Vec embeddings: https://huggingface.co/fse/word2vec-google-news-300
Use the GloVe embeddings: https://www.kaggle.com/datasets/gargvg/glovewordembeddings
Use the FastText embeddings: https://fasttext.cc/docs/en/english-vectors.html
(wiki-news-300d-1M-subword.vec.zip: 1 million word vectors trained with subword infomation on Wikipedia 2017, UMBC webbase corpus and statmt.org news dataset (16B tokens))
Word2Vec Tutorial: https://www.kaggle.com/code/pierremegret/gensim-word2vec-tutorial
All embeddings models Tutorial: https://radimrehurek.com/gensim/models/fasttext.html

Q1. Load the review column of the dataset and apply the following necessary preprocessing.

- Converting to lowercase
- Removing punctuation and numbers
- Tokenizing text
- Removing stopwords

Find the representation of each sentence using the word vectors of the words using all 3 models. Take 10000 samples from train.csv and 1000 samples from test.csv with stratified sampling so that the class distribution is maintained.

Q2. Consider the analogy examples such as and find the closest possible word

a) good − bad + excellent =
b) delivery − late + fast =

Take the analogy dataset from https://github.com/nicholas-leonard/word2vec/blob/master/questions-words.txt and check whether you get the correct analogy or not as shown in the above examples.

Compare all the embedding models and report their accuracies.

Q3. Use the same data that you created in Q1 and design a simple neural network with a single hidden layer and train your model. The input sentence representation is computed in the following two ways.

a. The sentence representation is the average of all the word vectors in the sentence.
b. The sentence representation is the weighted average of all the word vectors in the sentence where the weights are the TF-IDF scores of the words.

You need to design three neural networks corresponding to each of the embeddings. Compare the results in terms of macro F1-scores.